

Logarithmic mass of factorial prime-power blocks

Fixed valuation residue classes and a quantitative refinement

8 September 2026 Version 1.0

Abstract

For a fixed modulus, we give a proposed proof that the logarithmic mass of the prime-power blocks of a factorial is equally distributed among the residue classes of their valuations, also after restriction to a fixed interval on the logarithmic prime scale. The main argument uses Sander's prime exponential-sum estimate with fixed parameters, followed by a finite-band exhaustion. It yields an eventually periodic valuation-set formulation and, as an elementary consequence, finiteness of the positive integer solutions of $n! = x^2(x+1)$. A separate appendix gives a proposed uniform error $O_m(n(\log n)^{3/4}(\log \log \log n)^{1/4})$, using additional analytic arguments. The qualitative proof does not depend on that appendix.

Status and provenance. This is an AI-assisted working paper prepared for a public research record, not a peer-reviewed publication. Independent human mathematical verification and a comprehensive novelty check have not been completed. The results below are proposed deductions from the stated inputs; no claim of priority, third-party endorsement, or resolution of Erdős problem #398 is made.

1 Statement and context

Write $e_p(n) = v_p(n!)$ and $\mathcal{L}_n = \log(n!)$. All logarithms are natural and all sums indexed by p are over primes. For $m \geq 2$, define

$$M_{m,c}(n; a, b) = \sum_{\substack{n^a < p \leq n^b \\ e_p(n) \equiv c \pmod{m}}} e_p(n) \log p. \quad (1.1)$$

Thus each prime is weighted by the logarithm of its entire prime-power block in $n!$, not merely by its presence or by $\log p$.

Theorem 1.1 (Proposed qualitative mass formula). *Fix integers $m \geq 2$ and c , and real numbers $0 \leq a < b \leq 1$. Then*

$$M_{m,c}(n; a, b) = \left(\frac{b-a}{m} + o(1) \right) \mathcal{L}_n \quad (n \rightarrow \infty). \quad (1.2)$$

The limit is through all positive integers n , without averaging over n .

The proof in Sections 2–4 keeps all Fourier parameters fixed until after the limit in n . In particular it does not silently apply a fixed-dimensional equidistribution result in increasing dimension. Appendix A treats increasing parameters separately; neither Theorem 1.1 nor its consequences require that refinement.

The eventually periodic formulation in Section 5 matches the assertion formulated as Conjecture 5.1 in Berend–Harmse [1]. This is a comparison with that paper’s statement, not an assertion that the conjecture remains open today. The status of subsequent literature and of any equivalent existing result is unresolved in this working paper.

2 The fixed-parameter analytic input

Put $e(t) = \exp(2\pi it)$. To avoid confusing a two-variable expression with the von Mangoldt function, write

$$\mathcal{A}(U, V) = \left(\frac{\log U}{\log V} \right)^2, \quad \nu_5(J) = \frac{1}{J^5(\log(2J))^2}. \quad (2.1)$$

These are Sander’s $\Lambda(U, V)$ and $v_5(J)$.

We use Proposition 1 of Sander [2], printed p. 109, with the common notation on p. 97. Let $J > 8$, let $1 \leq j = j_1 < \dots < j_r \leq J$ be integers, and let h_i be real numbers with $h_1 \geq 1$ and $H = \max_i |h_i|$. The proposition gives absolute positive constants C, κ such that

$$\left| \sum_{p \leq N} e \left(x \sum_{i=1}^r h_i p^{-j_i} \right) \right| \leq C^J [N^{1-\kappa\nu_5(J)\mathcal{A}(N, xH)} + N^{(j+2)/2} x^{-1/2} + N^{5/6} H^2] \cdot (\log(xH))^{4J}, \quad (2.2)$$

provided

$$\exp(CJ(\log(2J))^3) \leq N \leq x^{1/j}. \quad (2.3)$$

The upper exponent is determined by the *smallest nonzero exponent* j , not by J . The convention that constants are absolute unless otherwise indicated is stated on printed p. 87 of [2].

Lemma 2.1. *Fix $R \geq 1$, a nonzero vector $h \in \mathbb{Z}^R$, and $m \geq 2$. Let $j = \min\{r : h_r \neq 0\}$. If $0 < \alpha < \beta < 1/j$ are fixed, then*

$$\sup_{n^\alpha \leq T \leq n^\beta} \frac{\log T}{T} \left| \sum_{p \leq T} e \left(\frac{n}{m} \sum_{r=1}^R h_r p^{-r} \right) \right| \rightarrow 0. \quad (2.4)$$

Proof. Delete zero coefficients, retaining their original exponents; gaps in the exponent set are permitted in (2.2). Take a complex conjugate if needed so that the first coefficient is a positive integer. Set $x = n/m$ and $J = \max(9, R)$. All these parameters other than x remain fixed. The lower condition in (2.3) holds eventually, and $\beta < 1/j$ gives $n^\beta \leq (n/m)^{1/j}$ eventually. Uniformly in the specified range, $\log T / \log(nH/m) \geq \alpha/2$ for large n . With $c_J = \kappa\nu_5(J) > 0$, the normalized right-hand side is bounded by

$$O_{R,h,m,\alpha,\beta} \left(\left[n^{-c_J\alpha^3/4} + n^{-(1-\beta j)/2} + n^{-\alpha/6} \right] (\log n)^{4J+1} \right) = o(1).$$

□

3 A fixed band and its logarithmic mass

Fix R and

$$\frac{1}{R+1} < \alpha < \beta < \frac{1}{R}. \quad (3.1)$$

For $n^\alpha < p \leq n^\beta$, Legendre's formula has exactly R nonzero terms:

$$e_p(n) = \sum_{r=1}^R \left\lfloor \frac{n}{p^r} \right\rfloor. \quad (3.2)$$

For *all* primes p , define a point on $\mathbb{T}^R = (\mathbb{R}/\mathbb{Z})^R$ by

$$z_{n,p} = \left(\left\{ \frac{n}{mp} \right\}, \dots, \left\{ \frac{n}{mp^R} \right\} \right).$$

For every fixed nonzero Fourier vector its smallest nonzero exponent satisfies $\beta < 1/R \leq 1/j$. Lemma 2.1 and the prime number theorem therefore give equidistribution of $z_{n,p}$ for $p \leq T$, uniformly for $n^\alpha \leq T \leq n^\beta$.

Here is the order of approximation implicit in this assertion. For a fixed box, choose continuous periodic upper and lower approximations with arbitrarily small difference of integrals. Approximate them uniformly by trigonometric polynomials and apply Lemma 2.1 to their finitely many modes. First let $n \rightarrow \infty$, then let the approximation error tend to zero. This also treats points on the box boundaries and keeps uniformity in T .

Let

$$g_c(z) = \mathbf{1}_{\{\sum_{r=1}^R \lfloor m z_r \rfloor \equiv c \pmod{m}\}}.$$

It is the indicator of a union of m^{R-1} half-open boxes, each of volume m^{-R} , so its integral is $1/m$. The encoding identity

$$\lfloor m\{y/m\} \rfloor \equiv \lfloor y \rfloor \pmod{m} \quad (3.3)$$

holds for every real y : write $y/m = q + \eta$ with $q \in \mathbb{Z}$ and $0 \leq \eta < 1$, and use $\lfloor y \rfloor = mq + \lfloor m\eta \rfloor$. Consequently $g_c(z_{n,p})$ equals the required valuation indicator inside the band, and

$$C_n(T) := \sum_{p \leq T} \left(g_c(z_{n,p}) - \frac{1}{m} \right) = o(T/\log T) \quad (3.4)$$

uniformly in the specified range. Outside the band g_c is only a proxy; the same proxy is used throughout this prefix sum.

Set $A = n^\alpha$, $B = n^\beta$ and $w_0(t) = \log t/t$. With $|C_n(t)| \leq \epsilon_n t/\log t$, $\epsilon_n \rightarrow 0$, partial summation gives

$$\begin{aligned} \sum_{A < p \leq B} \left(g_c(z_{n,p}) - \frac{1}{m} \right) \frac{\log p}{p} &= C_n(B)w_0(B) - C_n(A)w_0(A) - \int_A^B C_n(t)w_0'(t) dt \\ &= o(\log n). \end{aligned} \quad (3.5)$$

The boundary terms are $O(\epsilon_n)$ and the integral is $O(\epsilon_n \log(B/A))$. Also,

$$\sum_{A < p \leq B} \frac{\log p}{p} = (\beta - \alpha) \log n + o(\log n).$$

From (3.2), $e_p(n) = n/p + O_R(n/p^2 + 1)$. Chebyshev's bound $\vartheta(y) = \sum_{p \leq y} \log p \ll y$ implies

$$\sum_{A < p \leq B} \left| e_p(n) - \frac{n}{p} \right| \log p \ll_R n/A + B = o(n \log n).$$

Thus, since $\mathcal{L}_n \sim n \log n$,

$$M_{m,c}(n; \alpha, \beta) = \left(\frac{\beta - \alpha}{m} + o(1) \right) \mathcal{L}_n. \quad (3.6)$$

4 Exhausting the prime range

For each fixed $0 \leq t \leq 1$ the unrestricted mass satisfies

$$\sum_{p \leq n^t} e_p(n) \log p = (t + o(1)) \mathcal{L}_n. \quad (4.1)$$

For $t = 0$ the sum is empty. For $t > 0$, Legendre's formula gives $e_p(n) = n/(p-1) + O(\log n / \log p)$ uniformly for $p \leq n$. The sum of the weighted errors over $p \leq n^t$ is $O(\pi(n^t) \log n) = O_t(n^t)$. Moreover,

$$\sum_{p \leq y} \frac{\log p}{p-1} = \log y + o(\log y),$$

by the prime number theorem and partial summation. These observations prove (4.1), including $t = 1$.

Fix $\epsilon > 0$. Choose finitely many disjoint closed intervals $[\alpha_i, \beta_i] \subset (a, b)$, each strictly inside some reciprocal band $(1/(R_i + 1), 1/R_i)$, with total length $\ell > b - a - \epsilon$. To do so, first discard a short interval near zero, then small neighborhoods of the finitely many remaining reciprocal endpoints. All intervals and all R_i are fixed before n tends to infinity.

By (3.6), the residue- c mass on the selected bands is $(\ell/m + o(1)) \mathcal{L}_n$. By (4.1), their omitted total mass within $(n^a, n^b]$ is $(b - a - \ell + o(1)) \mathcal{L}_n$. Positivity gives

$$\frac{\ell}{m} \leq \liminf_n \frac{M_{m,c}(n; a, b)}{\mathcal{L}_n} \leq \limsup_n \frac{M_{m,c}(n; a, b)}{\mathcal{L}_n} \leq \frac{\ell}{m} + (b - a - \ell).$$

Letting $\epsilon \downarrow 0$ proves Theorem 1.1. In particular, this is an iterated-limit argument, not an application of (2.2) with increasing dimension.

5 Consequences and limitations

Corollary 5.1. *Let E be an eventually periodic set of positive integers, of natural density $\delta(E)$. For fixed $0 \leq a < b \leq 1$,*

$$\log \prod_{\substack{n^a < p \leq n^b \\ e_p(n) \in E}} p^{e_p(n)} = ((b - a)\delta(E) + o(1)) \mathcal{L}_n. \quad (5.1)$$

Proof. For a fixed positive valuation d , $\sum_{p: e_p(n)=d} e_p(n) \log p \leq d\vartheta(n) = O_d(n)$. A finite change in E therefore changes its mass by $O_E(n)$. After such a change, E is a union of residue classes for a fixed period. Sum (1.2) over those classes; period one follows directly from (4.1). $\square$

A finite union of arithmetic progressions in the positive integers is eventually periodic. With $a = 1 - \beta$, $b = 1 - \beta_1$, (5.1) has the form of [1, Conjecture 5.1, p. 1758].

Corollary 5.2. *Theorem 1.1 implies that $n! = x^2(x+1)$ has only finitely many solutions in positive integers n, x .*

Proof. Since $\gcd(x, x+1) = 1$, every prime dividing x has even valuation in $n!$. Hence $2 \log x \leq M_{2,0}(n; 0, 1)$. Along an unbounded sequence of solutions, $x \rightarrow \infty$ and

$$2 \log x = \left(\frac{2}{3} + o(1) \right) \mathcal{L}_n, \quad M_{2,0}(n; 0, 1) = \left(\frac{1}{2} + o(1) \right) \mathcal{L}_n,$$

a contradiction. Bounded n permits only finitely many positive x . $\square$

This supplies neither a numerical cutoff nor a list of all solutions. The example appears in [1, Example 5.1]; no novelty claim for its present-day finiteness status is made here.

What the statement does not say. It is not an equidistribution theorem for unweighted prime counts. For example, the fraction of primes $p \leq n$ with odd $e_p(n)$ tends to $\log 2$, not $1/2$. For fixed r and large n , one has $e_p(n) = r$ on $n/(r+1) < p \leq n/r$. The prime number theorem assigns that interval limiting proportion $1/(r(r+1))$. Truncate at $r \leq D$ and bound the remaining prime count by $\pi(n/(D+1))$; then let $D \rightarrow \infty$ to obtain $\sum_{r \text{ odd}} 1/(r(r+1)) = \log 2$. Such large primes contribute only $O(n)$, rather than order $n \log n$, to block mass.

The theorem does not resolve $n! + 1 = x^2$ (Erdős problem #398). In particular, for the related branch $n! = 4u^2(u^2 - 1)$, the mass constraint studied in the motivating calculations meets a leading coefficient of $1/2$ on both sides. Merely reducing the error in (1.2) does not turn that equality into a contradiction. No growing-modulus estimate is asserted either.

Separation from a known consequence. The finiteness of $n! = (u^2 - 1)(4u^2 - 3)^2$ is not offered as a new application. It already follows from [1, Theorem 4.1], using the divisor $Q(X) = (4X^2 - 3)^2$ of the degree-six polynomial on the right. Outside $p = 2, 3$, the primes at which Q has a root have density $1/2$, whereas $\deg Q/6 = 2/3$. This comparison does not require the present mass formula.

6 Verification status

The main argument's analytic dependency is precisely Sander's Proposition 1 with its common hypotheses. Internal checks included the smallest-exponent condition, proxy prefix sums, boundary approximation, partial summation, and the order of limits. The quantitative appendix has additional dependencies and a substantially larger verification burden.

The accompanying research repository contains finite arithmetic and Fourier tests. Such tests do not prove any analytic estimate, asymptotic statement, or parameter uniformity in this paper. AI-assisted reviews are not represented as independent human review. The full proof of the cited exponential-sum input has not been reproved here, and no exhaustive search for corrections or subsequent equivalent results has been completed. The present document is a research record with explicit proof claims and dependencies, not a certification of them.

A A proposed uniform quantitative refinement

This appendix is independent of the fixed-parameter limiting argument above. It records a stronger proposed estimate for preservation and inspection, not because the consequences in Section 5 need its precision. An earlier parameter-tracked route using Sander's Proposition 1 gives the candidate error $n(\log n)^{8/9}(\log \log n)^{1/3}$; the present argument instead returns to two intermediate analytic lemmas. No optimality is claimed.

Theorem A.1 (Proposed quantitative refinement). *For each fixed integer $m \geq 2$, uniformly for $c \pmod{m}$ and $0 \leq a < b \leq 1$,*

$$M_{m,c}(n; a, b) = \frac{b-a}{m} \mathcal{L}_n + O_m \left(n(\log n)^{3/4} (\log \log \log n)^{1/4} \right). \quad (\text{A.1})$$

Here n is sufficiently large in terms of m ; no numerical starting point or implied constant is supplied.

A.1 Parameters and bands

Put

$$L = \log n, \quad w = \log \log L, \quad K = \left\lfloor \rho_m (L/w)^{1/4} \right\rfloor, \quad d = \frac{1}{10K^2}, \quad (\text{A.2})$$

where $\rho_m > 0$ will be fixed sufficiently small. We may assume $K \geq 9$. For $1 \leq R \leq K$ let

$$\alpha_R = \frac{1}{R+1} + d, \quad \beta_R = \frac{1}{R} - d. \quad (\text{A.3})$$

These disjoint nonempty bands have total length

$$\lambda_K = \sum_{R=1}^K (\beta_R - \alpha_R) = 1 - \frac{1}{K+1} - \frac{1}{5K} = 1 - O(K^{-1}). \quad (\text{A.4})$$

Their nonemptiness follows even at $R = K$ from $1/(K(K+1)) > 1/(5K^2)$. Throughout,

$$\frac{K^2 \log L}{L} \rightarrow 0, \quad \frac{K^2 (\log(2K))^3}{L} \rightarrow 0, \quad \frac{L/K^3}{Kw} \asymp \rho_m^{-4}, \quad \frac{dL}{Kw} \rightarrow \infty. \quad (\text{A.5})$$

The lower bound on n may depend on fixed m and the chosen ρ_m , but not on bands, frequencies, or interval endpoints.

A.2 Symmetric Fourier smoothing

To distinguish smoothing widths from difference coefficients below, put

$$\omega = L^{-4}, \quad \delta = \omega/m, \quad H = \lceil m^2 L^8 \rceil. \quad (\text{A.6})$$

Let F_H be the nonnegative Fejér kernel of integral one and degree at most H , with coefficients bounded by one. Its tail satisfies

$$\epsilon_H := \int_{\|t\| \geq \delta} F_H(t) dt \ll (H\delta)^{-1} \ll_m L^{-4}.$$

On $\mathbb{T}^R$, set

$$g_{R,c}(z) = \mathbf{1}_{\{\sum_{r=1}^R \lfloor m z_r \rfloor \equiv c \pmod{m}\}}, \quad Q_{R,c} = g_{R,c} * F_H^{\otimes R}.$$

Then $0 \leq Q_{R,c} \leq 1$ and $\widehat{Q}_{R,c}(0) = 1/m$. If each coordinate is farther than δ from the grid $m^{-1}\mathbb{Z}$, all translations with coordinate distances less than δ preserve $g_{R,c}$. A union bound on the kernel tails therefore gives the pointwise estimate

$$|g_{R,c}(z) - Q_{R,c}(z)| \leq \sum_{r=1}^R \mathbf{1}_{\{\text{dist}(z_r, m^{-1}\mathbb{Z}) \leq \delta\}} + R\epsilon_H. \quad (\text{A.7})$$

This includes boundary points.

Let $\phi_a(z) = e(a[mz]/m)$. Finite character orthogonality gives

$$g_{R,c}(z) = \frac{1}{m} \sum_{a=0}^{m-1} e(-ac/m) \prod_{r=1}^R \phi_a(z_r).$$

Direct integration over the m cells yields

$$\widehat{\phi}_a(h) = m \mathbf{1}_{\{h \equiv a \pmod{m}\}} \int_0^{1/m} e(-hz) dz. \quad (\text{A.8})$$

Thus every nonconstant mode of $Q_{R,c}$ satisfies

$$h_1 \equiv \cdots \equiv h_R \equiv a \not\equiv 0 \pmod{m}. \quad (\text{A.9})$$

In particular *every coordinate is nonzero*. Its smallest exponent is one and its largest is R ; there is no need to remove zero coefficients. Taking a conjugate if necessary makes $h_1 \geq 1$. The bound $|\widehat{\phi}_a(h)| \ll_m \min(1, |h|^{-1})$ gives

$$\sum_{h \neq 0} |\widehat{Q}_{R,c}(h)| \leq [C_m(1 + \log H)]^R \leq \exp(A_m K w), \quad (\text{A.10})$$

since $\log(1 + \log H) = w + O_m(1)$.

For $z = z_{n,p}$ the boundary condition in (A.7) is exactly

$$\|n/p^r\| \leq m\delta = \omega. \quad (\text{A.11})$$

Here and below $\|\cdot\|$ denotes distance to the nearest integer. Identity (3.3) shows that $g_{R,c}(z_{n,p})$ is the true valuation indicator only on the R th retained band. Elsewhere it is again a consistently defined proxy.

A.3 The source of the losses in Proposition 1

It is important to distinguish the bound we use from a formal improvement of (2.2). On printed p. 103, Sander's Lemma 7 has the factor $\nu_5(j)$ and prefactor C^j , where j is the smallest exponent. The statement of Lemma 8 on the same page uses the weaker $\nu_5(J)$; its dyadic calculation on p. 104 still displays $\nu_5(j)$ before the conclusion. In the proof of Lemma 9, equation (70) on p. 105 likewise uses $\nu_5(J)$ after invoking Lemma 7. Since $j \leq J$ and ν_5 decreases, that weakening is valid but is not forced by a change of smallest exponent in the normalized phases used here.

The large logarithmic factors have a separate origin. Lemma 8 covers small M using the threshold $J^2 H^2 (3 \log(xH))^{2J}$ in (66). Lemma 9 covers the small- M range preceding (68), introducing a factor of order $(\log(xH))^{J+2}$. Lemma 11, including (72) on p. 108, then bounds these costs by the common factor $(\log(xH))^{4J}$. The proof below applies Lemma 7 only on intervals well above its thresholds and pays explicitly for discarded small intervals. It neither quotes Lemmas 8–9 nor replaces J by one in Proposition 1.

A.4 A prime exponential sum with smallest exponent one

Set $x = n/m$, $J = \max(9, R) \leq K$, and

$$f(v) = e \left(x \sum_{r=1}^R h_r v^{-r} \right), \quad h_1 \geq 1, \quad \max_r |h_r| \leq H.$$

We prove, uniformly for $n^{1/(K+1)} \leq T \leq n^{1-d}$,

$$\left| \sum_{p \leq T} f(p) \right| \ll L^{10} T \left[\exp \left(-c \frac{(\log T)^3}{L^2} \right) + \sqrt{T/x} + T^{-1/6} \right], \quad (\text{A.12})$$

where $c > 0$ and the implicit constant can be absolute. In particular, there is no logarithmic prefactor of degree depending on K .

Lemma 7 in the required form. For real coefficients $a_1 \geq 1$, $H' = \max_r |a_r|$, and $T' > 2$, the $j = 1$ case of [2, Lemma 7, p. 103] implies

$$\left| \sum_{P < u \leq P+P'} e \left(T' \sum_{r=1}^R a_r u^{-r} \right) \right| \ll P \exp \left(-c \frac{(\log P)^3}{\log^2(T'H')} \right) + \frac{P^2}{T'}, \quad (\text{A.13})$$

for $0 \leq P' < P$, provided

$$P > JH'(3 \log(T'H'))^J, \quad P > \exp(CJ(\log(2J))^3). \quad (\text{A.14})$$

The source's C^j and $\nu_5(j)$ are absolute constants at $j = 1$; we retain this instead of weakening C^j to C^J . The source includes its lower integer endpoint, whereas the form displayed here may remove one term. This $O(1)$ change is harmless in all applications below. There is *no* additional condition $T' > P$: the proof of Lemma 7 explicitly handles $P \geq T'$ by the term P^{j+1}/T' .

In our applications $H' \leq H$, $T'H' \leq xH$, and $P \geq \exp(L/(13(K+1)))$. Hence $\log(T'H') \leq 2L$, and (A.5) proves both conditions in (A.14) uniformly. The logarithms of their right-hand sides are respectively $O_m(K \log L + \log H)$ and $O(K(\log(2K))^3)$, both $o(L/K)$.

The Vaughan input and Type I. For

$$n^{1/(2(K+1))} \leq N \leq n^{1-d}, \quad (\text{A.15})$$

one has $N \leq x$ for sufficiently large n . Use [2, Lemma 10, p. 106] with $U_0 = V_0 = N^{1/3}$, rounded down if necessary. In the source's form of Vaughan's identity,

$$\left| \sum_{u \leq N} \Lambda(u) f(u) \right| \ll V_0 + (\log N) S_1 + |S_2|, \quad (\text{A.16})$$

where Λ now denotes the von Mangoldt function and

$$S_1 = \sum_{t \leq U_0 V_0} \max_{v > 0} \left| \sum_{v \leq s \leq N/t} f(st) \right|,$$

$$S_2 = \sum_{U_0 < u \leq N/V_0} \left(\sum_{\substack{d|u \\ d \leq U_0}} \mu(d) \right) \sum_{V_0 < v \leq N/u} \Lambda(v) f(uv).$$

The arithmetic coefficients in Type I are already accounted for by the external $\log N$ in (A.16); none is dropped here. Endpoint conventions or rounding add terms covered by the same estimates.

Put $M = N/t$. Discard $s \leq \sqrt{M}$ trivially and split the rest into dyadic intervals with lower endpoint $P \geq \sqrt{M}/2 \geq N^{1/6}/2$. The normalized phase is

$$f(st) = e \left(\frac{x}{t} \sum_{r=1}^R h_r t^{1-r} s^{-r} \right).$$

Its first coefficient is h_1 , its coefficient maximum is at most H , and $T' = x/t \geq N/t = M > 2$. Define

$$E_N = \exp \left(-c \frac{(\log N)^3}{L^2} \right),$$

decreasing the absolute c as necessary. Summing the interval lengths and their squares geometrically in (A.13) gives

$$\max_v \left| \sum_{v \leq s \leq M} f(st) \right| \ll ME_N + M^2 t/x + \sqrt{M}.$$

An arbitrary partial interval is a difference of prefixes with the same base. A full dyadic interval is covered by the source's strict length condition after removing an endpoint. For $c \leq 1/12$ and $\log N \leq L$, $E_N \geq N^{-1/12}$; thus $PE_N \geq N^{1/12}/2$ absorbs these $O(1)$ costs, also for the Type II applications below. Summing over t yields

$$S_1 \ll NE_N \log N + (N^2/x) \log N + N^{5/6}. \quad (\text{A.17})$$

Here $\sqrt{N} \sum_{t \leq N^{2/3}} t^{-1/2} \ll N^{5/6}$.

Type II. For $U_0 < M \leq N/V_0$, set

$$A(u) = \sum_{\substack{d|u \\ d \leq U_0}} \mu(d), \quad \mathcal{B}(u) = \sum_{V_0 < v \leq N/u} \Lambda(v) f(uv).$$

Elementary divisor counting gives

$$\sum_{M < u \leq 2M} |A(u)|^2 \ll M(\log N)^3. \quad (\text{A.18})$$

Indeed, enlarging to $u \leq 2M$ bounds the left side by $2M \sum_{d_1, d_2 \leq U_0} \gcd(d_1, d_2)/(d_1 d_2)$; grouping by the greatest common divisor gives three harmonic sums.

Expand $\sum_{M < u \leq M'} |\mathcal{B}(u)|^2$, where $M' \leq 2M$. The diagonal is $O(N(\log N)^2)$. For $2 \leq v_1 < v_2 \leq Y = N/M$ put

$$D_r(v_1, v_2) = v_1^{-r} - v_2^{-r} > 0.$$

Since $rt^{-r-1} \leq t^{-2}$ for $t \geq 2$, integration gives $0 < D_r/D_1 \leq 1$. The difference phase becomes

$$x \sum_r h_r D_r u^{-r} = (xD_1) \sum_r (h_r D_r / D_1) u^{-r}. \quad (\text{A.19})$$

Its first coefficient is again h_1 , and all coefficients are bounded by H . If $xD_1 > 2$, apply (A.13). If $xD_1 \leq 2$, the same bound follows trivially from $M^2/(xD_1) \geq M^2/2$. This also explains directly why a possible range $2 < xD_1 \leq M$ is harmless.

The identity $D_1^{-1} = v_1 v_2 / (v_2 - v_1)$ gives

$$\sum_{2 \leq v_1 < v_2 \leq Y} D_1^{-1} \ll Y^3 \log(2Y). \quad (\text{A.20})$$

Together with the $O(Y^2)$ pairs, this proves

$$\sum_{M < u \leq M'} |\mathcal{B}(u)|^2 \ll (\log N)^2 \left[N + \frac{N^2}{M} E_N + \frac{N^3 \log N}{Mx} \right]. \quad (\text{A.21})$$

Apply Cauchy–Schwarz with (A.18) and sum over $O(\log N)$ dyadic intervals. Since $\sqrt{MN} \ll N^{5/6}$, we get

$$|S_2| \ll L^5 \left[N E_N + N^{3/2} x^{-1/2} + N^{5/6} \right]. \quad (\text{A.22})$$

The halving of the saving exponent by Cauchy–Schwarz is incorporated in the absolute c defining E_N .

Removing the weights. Equations (A.16), (A.17), and (A.22), with $N^2/x \leq N^{3/2}x^{-1/2}$, yield

$$\left| \sum_{u \leq N} \Lambda(u) f(u) \right| \ll L^6 \left[N E_N + N^{3/2} x^{-1/2} + N^{5/6} \right]. \quad (\text{A.23})$$

Prime powers of exponent at least two cost $O(\sqrt{N} \log N)$. Remove $\log p$ by partial summation, splitting at $\sqrt{T}$. Below that point use a trivial Chebyshev bound; above it, (A.15) holds and $\log u \geq \frac{1}{2} \log T$, so only an absolute factor is lost in the saving exponent. The resulting logarithmic costs fit within L^{10} , proving (A.12). All constants used here are independent of R and the frequencies in their stated ranges.

A.5 Boundary errors counted over integers

Consider a retained-band interval

$$n^{\alpha_R} \leq P < V \leq \min(2P, n^{\beta_R}). \quad (\text{A.24})$$

The right endpoint is denoted V , distinct from the Vaughan parameter V_0 . We show, uniformly for $1 \leq r \leq R$,

$$\#\{k \in \mathbb{Z} : P < k \leq V, \|n/k^r\| \leq \omega\} \ll PL^{-4}. \quad (\text{A.25})$$

This bounds the corresponding prime count from above because the quantity being bounded is nonnegative.

For $1 \leq q \leq \lceil L^8 \rceil$ let

$$F = qnP^{-r}, \quad \theta = \frac{\log F}{\log P}, \quad f_r(u) = qnu^{-r}.$$

The band inequalities imply, for large n ,

$$\log P \geq L/(K+1), \quad F \geq \exp(dL), \quad 0 < \theta \leq K+2. \quad (\text{A.26})$$

For the middle assertion, $1 - r\beta_R \geq Rd \geq d$.

Large amplitude: $\theta \geq 3/4$. We apply [2, Lemma 2, p. 91]. For precision, the form needed here is as follows. Choose fixed $0 < c_3 < c_2 < c_1 < 1$ and $c_0 > 0$. For integer parameters N_s, P_s, P'_s, D, t with $0 \leq P'_s < P_s$,

$$P_s \geq 8^{1/c_3}, \quad D \geq \max\{12, (1 - c_1)^{-1/2}, (c_1 - c_2)^{-1/3}\}, \quad c_0 D \leq t < D, \quad (\text{A.27})$$

suppose the normalized derivative of order $D + 1$ has absolute value at most $P_s^{-c_1(D+1)}$ on $[N_s, N_s + 2P_s]$. Suppose also that for t distinct positive orders $s \leq D$, on $[N_s, N_s + P_s]$,

$$P_s^{-c_2 s} \leq |f^{(s)}(u)|/s! \leq P_s^{-c_3 s}.$$

The lemma bounds the exponential sum over $N_s \leq k \leq N_s + P'_s$ by

$$\ll P_s^{1-\gamma/D^2} (\log P_s)^{4/D^3}, \quad \gamma = \varrho c_0^2 \left(1 + \log \frac{3c_1}{\varrho c_0}\right)^{-2}, \quad (\text{A.28})$$

where $\varrho = \min\{c_3, c_1 - c_2 - 1/(D^2(D+1))\}$; the implicit constant is absolute. The selected derivatives in our application are positive, so their signed version also satisfies the source's hypotheses.

Choose

$$D = \lceil 32\theta \rceil + 12, \quad c_0 = 1/200, \quad c_1 = 15/16, \quad c_2 = 7/8, \quad c_3 = 1/8. \quad (\text{A.29})$$

Use the even orders $s \geq 2$ with $4\theta/3 \leq s \leq 4\theta$. Their number is at least $(\theta + 1)/4$: for $3/4 \leq \theta \leq 3/2$ the order two suffices, and above that range the interval-length estimate $4\theta/3 - 1 \geq (\theta + 1)/4$ applies. Since $D \leq 45(\theta + 1)$, the number t is at least $D/180 > D/200$; all selected orders are less than D , so $t < D$.

The exact derivative is

$$\frac{|f_r^{(s)}(u)|}{s!} = qn \binom{r+s-1}{s} u^{-r-s}. \quad (\text{A.30})$$

For $u \in [P, 4P]$ and $s \leq D + 1$, since $r, D = O(K)$,

$$\frac{\log(|f_r^{(s)}(u)|/s!)}{\log P} = \theta - s + O(K/\log P) = \theta - s + o(1) \quad (\text{A.31})$$

uniformly. Here $\binom{r+s-1}{s} \leq 2^{r+s-1}$ and (A.5) were used. The chosen orders satisfy $1/4 \leq \theta/s \leq 3/4$, and their derivatives are positive. The margins in the exponents therefore give, for all sufficiently large n ,

$$P^{-7s/8} \leq f_r^{(s)}(u)/s! \leq P^{-s/8}, \quad |f_r^{(D+1)}(u)|/(D+1)! \leq P^{-15(D+1)/16}. \quad (\text{A.32})$$

For the last inequality use $\theta/(D+1) \leq 1/32$. We have $D \geq 36$, all lower conditions in (A.27) hold, and $\varrho \geq 1/32$. Thus $\gamma \geq \gamma_* > 0$ for an absolute constant independent of K, r, q .

To match integer endpoints, if the sum is nonempty take $N_s = P_s = \lfloor P \rfloor + 1$ and $P'_s = \lfloor V \rfloor - N_s$. Then $0 \leq P'_s < P_s$ and the required derivative domains are contained in $[P, 4P]$ for large P . Replacing P by $P_s = P + O(1)$ preserves (A.32) by its strict exponent margins. Consequently

$$\left| \sum_{P < k \leq V} e(qn/k^r) \right| \ll P(\log P)^4 \exp(-c \log P/D^2) \ll PL^4 \exp(-cL/K^3). \quad (\text{A.33})$$

Small amplitude: $\theta < 3/4$. Here f'_r is monotone and

$$\sup_{[P, 2P]} |f'_r| \leq rF/P \leq KP^{-1/4} < 1/4, \quad \inf_{[P, 2P]} |f'_r| \geq \frac{rF}{2^{r+1}P}.$$

The sum–integral comparison and first-derivative integral bound in [2, Lemmas 4–5, pp. 101–102] yield

$$\left| \sum_{P < k \leq V} e(qn/k^r) \right| \ll \frac{2^r P}{rF} + 1 \ll P \exp(CK - dL) + 1. \quad (\text{A.34})$$

Because $L/K^3 \gg \log L$ and $dL \gg K + \log L$, (A.33)–(A.34) imply, for every fixed $A > 0$,

$$\max_{1 \leq q \leq \lceil L^8 \rceil} \left| \sum_{P < k \leq V} e(qn/k^r) \right| \ll_A PL^{-A}. \quad (\text{A.35})$$

The lower threshold for n here may also depend on A . This is a bound in terms of the parent scale P , not the potentially much shorter length $V - P$.

A one-dimensional majorant. Set $H_b = \lceil L^8 \rceil$ and

$$\mathcal{W}(y) = \mathbf{1}_{\{\|y\| \leq 2\omega\}} * F_{H_b}(y) + \eta, \quad \eta = C/(H_b\omega).$$

For a sufficiently large absolute C , the tail estimate for the Fejér kernel gives, including boundary points,

$$\mathbf{1}_{\{\|y\| \leq \omega\}} \leq \mathcal{W}(y), \quad \widehat{\mathcal{W}}(0) \ll \omega, \quad \sum_{q \neq 0} |\widehat{\mathcal{W}}(q)| \ll \log L.$$

Apply (A.35) with $A = 10$ (negative frequencies follow by conjugation). The result is $\sum_{P < k \leq V} \mathcal{W}(n/k^r) \ll P\omega + PL^{-10} \log L \ll PL^{-4}$, which proves (A.25). In particular,

$$\sum_{P < p \leq V} |g_{R,c}(z_{n,p}) - Q_{R,c}(z_{n,p})| \ll_m KPL^{-4}. \quad (\text{A.36})$$

Only this nonnegative error has been enlarged from primes to integers; the oscillatory main Fourier sum has not.

A.6 Combining costs and savings

From (A.10) and (A.12), the positive cost in the main Fourier sum is at most $\exp(A_m Kw + 10 \log L + O(1))$. For $T \geq n^{1/(K+1)}$ its first negative exponent is at least cL/K^3 , after an absolute adjustment of c . Choose ρ_m small enough that $c\rho_m^{-4} > 2A_m + 10$. Since $K^4 w \leq \rho_m^4 L$ and $\log L = o(Kw)$, the saving absorbs the full positive cost with room for any fixed negative power of L . The other terms have savings $\sqrt{T/x} \leq \sqrt{m} \exp(-dL/2)$ and $T^{-1/6} \leq \exp(-L/(6(K+1)))$, which are still stronger. Thus

$$\left| \sum_{p \leq T} \left(Q_{R,c}(z_{n,p}) - \frac{1}{m} \right) \right| \ll_m TL^{-10} \quad \left(n^{1/(K+1)} \leq T \leq n^{1-d} \right). \quad (\text{A.37})$$

Subtract two such prefixes and use (A.36). On every interval (A.24),

$$\left| \sum_{P < p \leq V} \left(\mathbf{1}_{\{e_p(n) \equiv c \pmod{m}\}} - \frac{1}{m} \right) \right| \ll_m KPL^{-4} \ll_m \frac{P}{K^3 \log P}. \quad (\text{A.38})$$

For the last inequality, $K^4 \log P/L^4 \leq K^4/L^3 \rightarrow 0$.

A.7 Weights and completion of the estimate

For $\alpha_R \leq s < t \leq \beta_R$, split $(n^s, n^t]$ into ratio-at-most-two intervals. Partial summation of (A.38) with weight $\log p/p$ costs $O_m(K^{-3})$ per interval. There are $O((t-s)L+1)$ such intervals, including a possible short final one. Hence

$$\left| \sum_{n^s < p \leq n^t} \left(\mathbf{1}_{\{e_p(n) \equiv c \pmod{m}\}} - \frac{1}{m} \right) \frac{\log p}{p} \right| \ll_m K^{-3}((t-s)L+1). \quad (\text{A.39})$$

We use the elementary uniform estimate

$$\sum_{p \leq y} \frac{\log p}{p} = \log y + O(1) \quad (y \geq 2). \quad (\text{A.40})$$

For completeness, the factorial identity and Chebyshev's bound give

$$\log(N!) = \sum_{d \leq N} \Lambda(d) \lfloor N/d \rfloor, \quad \sum_{d \leq N} \frac{\Lambda(d)}{d} = \log N + O(1).$$

For the second equality the floor errors total $O(N)$, since $\sum_{d \leq N} \Lambda(d) = O(N)$. Removing prime powers of exponent at least two costs $O(1)$, proving (A.40). Rounding extends it to real y .

On a retained band, Legendre's formula gives

$$\left| e_p(n) - \frac{n}{p} \right| \leq \frac{2n}{p^2} + R, \quad \sum_{n^s < p \leq n^t} \left| e_p(n) - \frac{n}{p} \right| \log p \ll n^{1-s} + Kn^t.$$

There is no extra $\log n$ here: partial summation of $\vartheta(y) \ll y$ gives $\sum_{p > A} (\log p)/p^2 \ll 1/A$. Consequently the mass on the subband is

$$\frac{t-s}{m} nL + O_m(nK^{-3}((t-s)L+1) + n + n^{1-s} + Kn^t). \quad (\text{A.41})$$

Intersect $(a, b]$ with the retained bands, and let ℓ be the sum of the resulting lengths. Summing (A.41) over at most K pieces gives

$$\frac{\ell}{m} nL + O_m(nL/K^3 + Kn). \quad (\text{A.42})$$

Indeed, the last two remainder terms total at most $O(Kn \exp(-L/(K+1)) + K^2 n \exp(-dL)) = o(n)$. The same elementary estimates without the residue restriction give total retained mass $\lambda_K nL + O(Kn)$. Since $\mathcal{L}_n = nL + O(n)$ and $K^2 \leq L$, the nonnegative omitted mass is bounded by

$$0 \leq M_{\text{omit}} \ll nL/K. \quad (\text{A.43})$$

Also $0 \leq b - a - \ell \leq 1 - \lambda_K = O(K^{-1})$. Adding the omitted contribution to (A.42) proves $M_{m,c}(n; a, b) = (b-a)nL/m + O_m(nL/K)$, uniformly in a, b, c . Finally,

$$nL/K \asymp_m nL^{3/4} w^{1/4} = n(\log n)^{3/4} (\log \log \log n)^{1/4},$$

which proves the proposed estimate (A.1) from the stated inputs.

A.8 Scope and remaining verification burden

The balance behind the three logarithms is $L/K^3 \gtrsim K \log \log L$. This is an asymptotic parameter calculation, not a claim of useful numerical savings at accessible values of n . The dominant error is the omitted mass of logarithmic length $O(1/K)$, not the finer discrepancy in (A.38). The estimate guarantees relative error tending to zero when

$$(b - a)K \rightarrow \infty, \quad \text{equivalently} \quad \frac{b - a}{L^{-1/4}(\log \log L)^{1/4}} \rightarrow \infty.$$

It remains an absolute estimate on shorter intervals but does not show that their main terms dominate.

No endpoint-uniform error can be $o_m(n)$. Take $a = 1 - \log 2/L$, $b = 1$, and $c = 0 \pmod{m}$. All primes in $(n/2, n]$ have $e_p(n) = 1$, so the true mass is zero while the proposed main term is $\log 2 \mathcal{L}_n/(mL) \sim (\log 2/m)n$. This lower bound neither establishes nor rules out an $O_m(n)$ upper bound.

The priority checks for this appendix are the uniform use of Lemma 7 and the Vaughan calculation in Section A.4, the derivative hypotheses of Lemma 2 in Section A.5, and the transition from the nonnegative boundary bound to weighted mass. Source statements and these calculations have received internal AI-assisted checks, not independent human certification. This appendix introduces no new Diophantine consequence beyond those already supplied by the qualitative formula in this paper.

References

- [1] D. Berend and J. E. Harmse, *On Polynomial-Factorial Diophantine Equations*, Transactions of the American Mathematical Society **358** (2006), 1741–1779. doi:10.1090/S0002-9947-05-03780-3.
- [2] J. W. Sander, *On the order of prime powers dividing $\binom{2n}{n}$* , Acta Mathematica **174** (1995), 85–118. doi:10.1007/BF02392802.